

SUMMARY

Home Assistant + MQTT Integration — Nakshatra Automation

Name: Monish M

Register Number: 42110819

MQTT Topic Used: home/monishm42110819-2025/sensor

Extra Sensor: Light Sensor (digital 0 = OFF, 1 = ON)

Explanation of Extra Sensor

I used a simulated digital Light sensor that switches between ON/OFF states. Since no physical sensor was connected, I simulated realistic smart-home behavior where light status changes based on activity.

Steps Followed (in my own words)

1. Installed Home Assistant on macOS using Docker and completed the initial setup.
2. Installed Mosquitto MQTT broker using Homebrew and verified broker functionality with `mosquitto_sub` and `mosquitto_pub`.
3. Wrote a Python script that publishes JSON data (temperature, humidity, light) along with my name and register number to my unique MQTT topic.
4. Configured Home Assistant's `mqtt.sensor` entries in `configuration.yaml` to read and display values from my MQTT topic.
5. Added my three sensors to the Home Assistant dashboard.
6. Verified live updates from the Python script into MQTT and Home Assistant.
7. Recorded the final 2–4 minute video with face verification, timestamp, MQTT output, Python script, and Home Assistant dashboard.
8. Took mandatory screenshots and uploaded all files to GitHub as required.